

Sabin Baral

Hattiesburg, MS | (601) 307-2765 | Sabin.Baral@usm.edu | [LinkedIn](#) | [GitHub](#) | [Portfolio](#)

Undergraduate Researcher in Polymer Science | High-Throughput Experimentation and Closed-Loop Materials Characterization

EDUCATION

The University of Southern Mississippi

Hattiesburg, MS

Bachelor of Science in Polymer Science and Engineering | Minor: Chemistry | Honors Scholar

May 2028

Relevant Coursework: Polymer Rheology, Polymer Mechanics, Polymer Processing, Calculus III, Differential Equations

GPA: 4.0/4.0 (President's List, Fall 2024 – Spring 2026)

PUBLICATIONS

Upreti, S.; Xu, L.; Moniruzzaman, M.; Wang, Y.; Adhikari, K.; Baral, S.; Patton, D.; Ma, B.; Xu, J.; Li, R.; Zhu, C.; Xia, W.; Gu, X. "[A Robotic High-Throughput Grid-Search Platform for Mapping Phase Behavior in Triblock Copolymer–Homopolymer Blends](#)." *ACS Nano*.

EXPERIENCE

Undergraduate Researcher and Instrumentation Engineer

Jul 2025 – Present

Gu Research Group

Hattiesburg, MS

- Developed a Python hardware abstraction layer (HAL) integrating robotic arms, motorized stages, and spectrometers into a closed-loop materials lab
- Fabricated and AFM-characterized **1,000+** thin-film samples, improving throughput by **40%**
- Performed **GISAXS/GIWAXS** experiments at **ALS, NSLS-II**, and in-house on 100+ samples
- Designed and fabricated custom sample holders and instrument housings, increasing storage capacity by **500%**

US CMS PURSUE Research Intern

May 2026 – Jul 2026

Fermi National Accelerator Laboratory (Fermilab) & Kansas State University

Batavia, IL & Manhattan, KS

- Cleanroom-tested **50+** TFPX bare modules, calibrating VDD/IREF and validating silicon stability up to 110 V
- Executed automated pixel calibration (PixelAlive, threshold equalization, noise mapping) using **Ph2_ACF**
- Built a Python GUI for CMS Read-Out Chip testing, reducing setup time by **75%** and eliminating terminal-based workflows

VeriScript – AI Transcript Fraud Detection Platform

Mar 2026 – May 2026

Mississippi State Board of Nursing

Jackson, MS | [GitHub](#)

- Designed a serverless pipeline (**S3→Lambda→Step Functions**) for automated transcript processing
- Built a **14-rule** compliance engine and deployed a React/TypeScript dashboard deployed via **AWS CDK**

Visiting Research Scholar

Sep 2025

Lawrence Berkeley National Laboratory

Berkeley, CA

- Used automated AFM to characterize phase behavior across **500+** polymer thin-film samples for a pre-screening database
- Assembled and optimized a robotic spin-coating and thermal annealing system and authored operating protocols

Undergraduate Research Assistant

Oct 2024 – May 2025

Center for Optoelectronics and Devices

Hattiesburg, MS

- Fabricated organic solar cells (**15–18%** efficiency) via spin coating, thermal evaporation, and encapsulation techniques
- Characterized device performance and stability through AFM, TGA, and DSC to investigate degradation mechanisms
- Correlated electrical performance with morphological, thermal, and structural changes to evaluate long-term reliability

SKILLS

- Materials Characterization:** GISAXS/GIWAXS, AFM, DLS, DSC, TGA, FTIR, UV-Vis, Cleanroom Probe Stations
- Scientific Computing:** Python, MATLAB, Ph2_ACF, CMSITminiDAQ
- Cloud & AI Tools:** AWS (Lambda, Step Functions, DynamoDB, CDK, S3), Amazon Bedrock (Nova Pro), Textract
- Design, Prototyping & OS:** AutoCAD, OpenSCAD, Cura, Bambu Lab H2D, Linux (Pop!_OS, AlmaLinux, CentOS)

HONORS AND AWARDS

Honors Keystone Award, USM

Jul 2026

Winner, RoboSoccer Tournament – Google Developer Student Clubs, USM

Sep 2025

Valedictorian and Student Body President Scholarship, USM

Jul 2024